NLP (Natural language processing)

Tokenization

My name is Mehmed. ? !

token

Morfolojik gitmek istersek (şekle göre)

Benim adım Mehmed.

Ben-im ad-ım

kelimelere bölerek word piece

→ Morfolojik gramere göre

Türkçede 10000 kelime

10 000

ahmed → 2000

Kare not encoding

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Elimizde dokümanlar var

	D_1	D_2	- - -
D_1		3	5

↙ TF

Term frequency

Tüm dokümanlarda iki kelime aynı frekansla geçmemeli

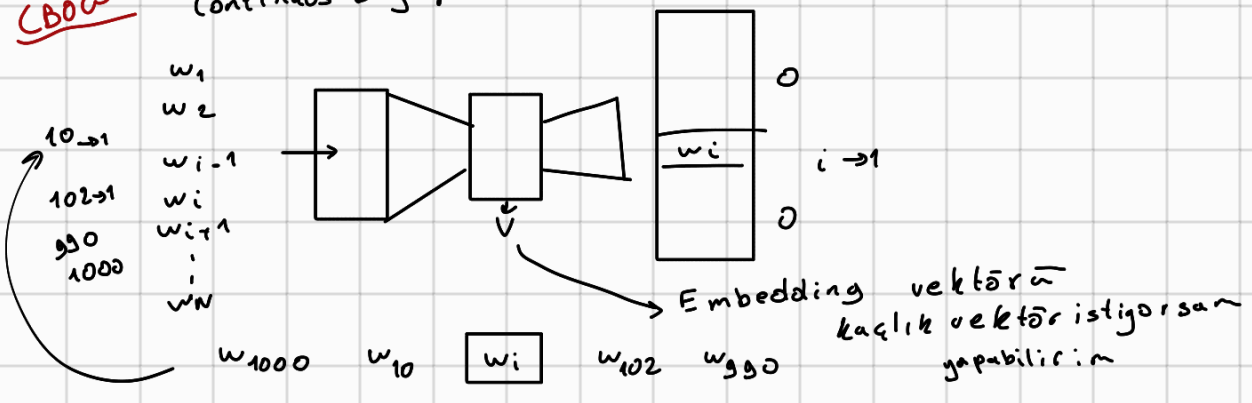
Bow

TF

TF-IDF → baskınlığı azaltmak için

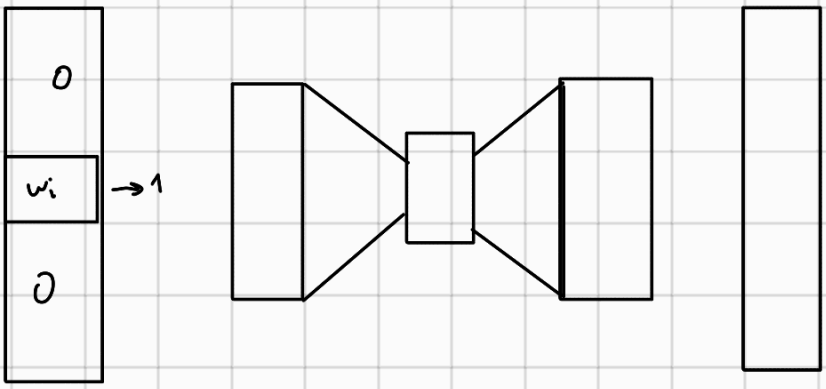
Word 2Vec

CBOW (Continuous Bag of Words)



Skip gram / n-gram 2gram iki komşuluk

tersten de yapabilirim



Global vectors for word Representation (Glove)

$$J = \sum_{ij} f(x_{ij}) \left(w_i^T \tilde{w}_j + b_i + b_j - \log X_{ij} \right)^2$$

w_i^T → bunu bulmaya çalışıyoruz kelimenin ağırlığı

X_{ij} → i'inci doküman j'inci kelimenin geçme frekansı

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apple = $t_1 + t_2$

app/le → subword

word embedding

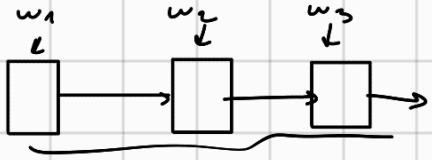
t_1 t_2

Fast Text

Static embedding

Covid geldi yeni kelime

Benim adım Mehmet / Seq 2Seq ↙ Supervized



encoder

RNN → exploding vanishing

unutma fazla hafızaya

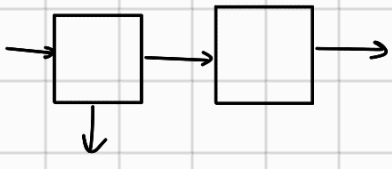
My name is Mehmet

$\tilde{w}_1$ $\tilde{w}_2$ $\tilde{w}_3$ $\tilde{w}_4$

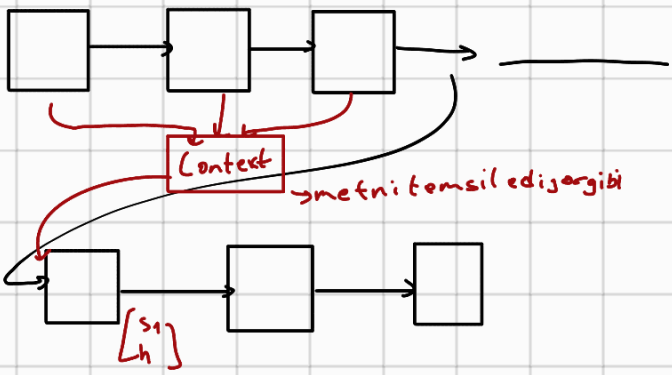
sonraki güncellemede LSTM

Bu metin kısa olmadıktan sonra context yakalayamıyor.

decoder



Attention Mechanism

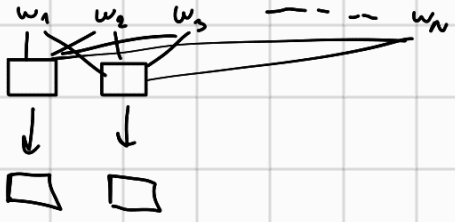


kelimeleri hiden layerle carpiyoruz

softmax $\begin{cases} w.h \\ [\alpha_1, \alpha_2, \dots \alpha_{100}] \end{cases}$

hangi kelime ne kadar önemli.

Sonra dönmekteyine MLP koyuyoruz



paralelde calişabiliyor

Encoding :

input encoding

Multi-Head Attention

Feed forward

Decoder:

output embedding

Multihead Attention

Multihead Attention

Feed Forward

GPT sadece decoder

Dynamic Embedding

2019

BERT: Sadece Encoder, kelimeleri ayirabiliyor

BERT NLP'nin golosu

BERT : Bidirectional Encoder Representions from Transformers

ali topu at

/a/l/i/ / /t/o/p/u/ /a/t/

arada hatalar, düzelting için token sayiyor

ELMO: Embeddings from Language model

context meknizmsiyon

